

Scatter-Regularized Cross-Multi-Kernel Contrastive Learning for One-Class Anomaly Detection

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Abstract

We present the methodology of a scatter-regularized cross-multi-kernel contrastive framework (SCMK) for one-class anomaly detection. The full manuscript is in preparation; this file contains the Methodology section.

Keywords: anomaly detection, multiple kernel learning, contrastive learning, one-class classification, representation learning

1. Methodology

1.1. Problem formulation and notation

We address *one-class* (semi-supervised) anomaly detection. Let $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N$ with $\mathbf{x}_i \in \mathbb{R}^D$ denote the data after preprocessing,¹ and let $y_i \in \{0, 1\}$ be the latent label, where $y_i = 0$ marks a normal instance and $y_i = 1$ an anomaly. Only the normal subset $\mathcal{X}_n = \{\mathbf{x}_i : y_i = 0\}$, of size $N_n = |\mathcal{X}_n|$, is available at training time; anomalies are neither observed nor simulated. The goal is to learn a scoring function $a : \mathbb{R}^D \rightarrow \mathbb{R}$ such that $a(\mathbf{x})$ is large for anomalies and small for normal points.

Our framework, termed *scatter-regularized cross-multi-kernel* contrastive learning (SCMK), couples a multiple-kernel contrastive encoder with a one-class detection head. It comprises four components, summarised in Algorithm 1: (i) a multi-scale kernel bank that injects geometric priors at several resolutions (Section 1.2); (ii) a cross-kernel contrastive objective that aligns the views produced by different kernels (Section 1.3); (iii) a multi-kernel

¹Nominal attributes are one-hot encoded and numerical attributes are min-max scaled to $[0, 1]$, so that all feature dimensions share a comparable range.

scatter regularizer that explicitly compacts the normal class in every kernel-induced embedding (Section 1.4); and (iv) a linear one-class support vector machine (OC-SVM) that scores anomalies in the concatenated representation (Section 1.6).

1.2. Multi-scale kernel bank

To endow the encoder with multi-resolution sensitivity, we instantiate a bank of K Gaussian kernels whose bandwidths are anchored to the intrinsic scale of the normal data. Let

$$\text{med} = \text{median}\{\|\mathbf{x}_i - \mathbf{x}_j\|_2 : \mathbf{x}_i, \mathbf{x}_j \in \mathcal{X}_n, i \neq j\} \quad (1)$$

be the median pairwise Euclidean distance estimated *from normal samples only*, which prevents contamination of the scale statistic by outliers. The k -th kernel is

$$\kappa_k(\mathbf{a}, \mathbf{b}) = \exp\left(-\frac{\|\mathbf{a} - \mathbf{b}\|_2^2}{2t_k^2}\right), \quad t_k = r_k \cdot \text{med}, \quad (2)$$

with multiplicative ratios $\{r_k\}_{k=1}^K = \{0.1, 0.5, 1.0, 2.0, 5.0\}$ ($K = 5$). Small ratios produce sharp kernels that resolve fine-grained local neighbourhoods, whereas large ratios produce broad kernels that capture the global manifold geometry. The bank therefore provides K complementary similarity measures that the contrastive stage will reconcile.

1.3. Cross-kernel contrastive representation learning

Per-kernel embeddings. For each kernel we attach an independent, bias-free linear projection head $g_k(\mathbf{x}) = \mathbf{W}_k \mathbf{x}$ with $\mathbf{W}_k \in \mathbb{R}^{d \times D}$, followed by ℓ_2 normalization,

$$\mathbf{h}_{k,i} = \frac{\mathbf{W}_k \mathbf{x}_i}{\|\mathbf{W}_k \mathbf{x}_i\|_2} \in \mathbb{S}^{d-1}, \quad k = 1, \dots, K, \quad (3)$$

so that every embedding lives on the unit hypersphere $\mathbb{S}^{d-1}$. Removing the bias keeps the representation centred at the origin, which is consistent with the origin-separating hyperplane assumption of the downstream OC-SVM, while the normalization equalises the scale across kernels.

Cross-kernel positive pairs.. Whereas conventional contrastive learning forms positive pairs through stochastic data augmentation, we treat the *kernel index as the view*: the two embeddings $\mathbf{h}_{k,i}$ and $\mathbf{h}_{l,i}$ of the *same* sample under *different* kernels constitute a positive pair, and embeddings of distinct samples are negatives. Consider a mini-batch of B normal samples and a kernel pair (k, l) . Stacking the two views yields $\mathbf{F}^{(kl)} = [\mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,B}, \mathbf{h}_{l,1}, \dots, \mathbf{h}_{l,B}]^\top \in \mathbb{R}^{2B \times d}$, whose rows are indexed by $m, n \in \{1, \dots, 2B\}$. The pairwise affinity is the *symmetrised kernel response* evaluated on the embeddings,

$$s_{mn}^{(kl)} = \frac{1}{2} [\kappa_k(\mathbf{f}_m, \mathbf{f}_n) + \kappa_l(\mathbf{f}_m, \mathbf{f}_n)]. \quad (4)$$

Because the embeddings are unit-norm, the Gaussian response reduces to a monotone function of cosine similarity, $\kappa_k(\mathbf{f}_m, \mathbf{f}_n) = \exp(-(1 - \langle \mathbf{f}_m, \mathbf{f}_n \rangle) / t_k^2)$, so t_k^2 plays the role of a contrastive temperature that is automatically adapted to the data scale through Eq. (1).

Cross-kernel InfoNCE.. For anchor n , let $\pi(n)$ index its cross-view positive (i.e. $\pi(n) = n + B$ for $n \leq B$ and $\pi(n) = n - B$ otherwise). The InfoNCE loss for the pair (k, l) is

$$\mathcal{L}_{\text{con}}^{(kl)} = -\frac{1}{2B} \sum_{n=1}^{2B} \log \frac{\exp(s_{n,\pi(n)}^{(kl)})}{\sum_{m \neq n} \exp(s_{nm}^{(kl)})}, \quad (5)$$

where the denominator excludes the self-similarity term $m = n$. Averaging over all $\binom{K}{2}$ unordered kernel pairs gives the cross-kernel contrastive objective

$$\mathcal{L}_{\text{con}} = \frac{1}{\binom{K}{2}} \sum_{1 \leq k < l \leq K} \mathcal{L}_{\text{con}}^{(kl)}. \quad (6)$$

Minimising Eq. (6) drives the encoder to a *view-invariant* representation in which all kernels agree on the embedding of each normal sample, i.e. $\mathbf{h}_{k,i} \approx \mathbf{h}_{l,i}$, while keeping distinct samples mutually discriminable through the negative terms.

1.4. Multi-kernel scatter regularization

Motivation.. The contrastive objective (6) enforces *cross-kernel consistency* but does not constrain the *absolute compactness* of the normal class: it is invariant to a rotation that spreads normal embeddings across the sphere as

long as the per-sample views remain aligned. A one-class detector, however, benefits directly from a tightly concentrated normal region. We therefore introduce a regularizer that minimises the dispersion of the normal class within every kernel-induced embedding space, drawing on the one-class specialisation of centroid multiple-kernel k -means (CMKKM).

Definition.. Let $\boldsymbol{\mu}_k = \frac{1}{B} \sum_{i=1}^B \mathbf{h}_{k,i}$ be the empirical centroid of the normal embeddings under kernel k . The multi-kernel scatter penalty is

$$\mathcal{L}_{\text{scat}} = -\frac{1}{K} \sum_{k=1}^K \|\boldsymbol{\mu}_k\|_2^2. \quad (7)$$

It is computed in $O(KBd)$ time — a single mean per kernel — and thus adds negligible overhead.

Exact equivalence to within-class scatter.. The next result shows that Eq. (7) is not a heuristic surrogate but is *exactly* (up to an additive constant) the average within-class scatter of the normal embeddings.

Proposition 1 (Scatter equivalence). *For unit-norm embeddings $\mathbf{h}_{k,i} \in \mathbb{S}^{d-1}$, define the average normalised within-class scatter $\sigma^2 = \frac{1}{NK} \sum_{k=1}^K \sum_{i=1}^N \|\mathbf{h}_{k,i} - \boldsymbol{\mu}_k\|_2^2$. Then*

$$\sigma^2 = 1 + \mathcal{L}_{\text{scat}}. \quad (8)$$

Consequently, minimising $\mathcal{L}_{\text{scat}}$ is equivalent to minimising the within-class scatter σ^2 , and equivalently to maximising the mean intra-class kernel alignment $\frac{1}{NK} \sum_k \sum_{i,j} \langle \mathbf{h}_{k,i}, \mathbf{h}_{k,j} \rangle = \frac{1}{K} \sum_k \|\boldsymbol{\mu}_k\|_2^2$.

Proof. Fix a kernel k . Expanding the squared norm and using $\|\mathbf{h}_{k,i}\|_2^2 = 1$,

$$\begin{aligned} \sum_{i=1}^N \|\mathbf{h}_{k,i} - \boldsymbol{\mu}_k\|_2^2 &= \sum_{i=1}^N \left(\|\mathbf{h}_{k,i}\|_2^2 - 2 \langle \mathbf{h}_{k,i}, \boldsymbol{\mu}_k \rangle + \|\boldsymbol{\mu}_k\|_2^2 \right) \\ &= N - 2N \|\boldsymbol{\mu}_k\|_2^2 + N \|\boldsymbol{\mu}_k\|_2^2 = N(1 - \|\boldsymbol{\mu}_k\|_2^2), \end{aligned} \quad (9)$$

where $\sum_i \langle \mathbf{h}_{k,i}, \boldsymbol{\mu}_k \rangle = N \langle \boldsymbol{\mu}_k, \boldsymbol{\mu}_k \rangle = N \|\boldsymbol{\mu}_k\|_2^2$ follows from the definition of $\boldsymbol{\mu}_k$. Averaging Eq. (9) over k and dividing by N yields $\sigma^2 = 1 - \frac{1}{K} \sum_k \|\boldsymbol{\mu}_k\|_2^2 = 1 + \mathcal{L}_{\text{scat}}$. The alignment identity is immediate from $\sum_{i,j} \langle \mathbf{h}_{k,i}, \mathbf{h}_{k,j} \rangle = N^2 \|\boldsymbol{\mu}_k\|_2^2$. $\square$

Remark 1 (Connection to one-class CMKKM). *The one-class instance of centroid multiple-kernel k -means maximises $\text{Tr}(\mathbf{H}^\top \mathbf{K}_\eta \mathbf{H})$ with a single soft cluster indicator $\mathbf{H} = N^{-1/2} \mathbf{1}$ and a convex kernel combination $\mathbf{K}_\eta = \sum_k \eta_k \Phi_k \Phi_k^\top$, where $\Phi_k = [\mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,N}]^\top$ realises a linear kernel on the embeddings. A short calculation gives $\text{Tr}(\mathbf{H}^\top \mathbf{K}_\eta \mathbf{H}) = N \sum_k \eta_k \|\boldsymbol{\mu}_k\|_2^2$, which under uniform weights $\eta_k = 1/K$ equals $-N \mathcal{L}_{\text{scat}}$. Hence $\mathcal{L}_{\text{scat}}$ is precisely the (negated, scale-normalised) one-class CMKKM compactness objective, providing a multiple-kernel-learning justification for the penalty.*

1.5. Joint objective and collapse avoidance

The encoder $\{\mathbf{W}_k\}_{k=1}^K$ is trained by minimising the combined loss

$$\mathcal{L} = \mathcal{L}_{\text{con}} + \lambda \mathcal{L}_{\text{scat}}, \quad (10)$$

where $\lambda \geq 0$ trades off cross-kernel consistency against intra-class compactness; $\lambda = 0$ recovers the purely contrastive backbone.

A natural concern is representational *collapse*: the scatter term alone is trivially minimised ($\mathcal{L}_{\text{scat}} = -1$) when all embeddings of a kernel coincide ($\|\boldsymbol{\mu}_k\|_2 = 1$). This degenerate solution is, however, suppressed by the contrastive term. In the alignment–uniformity decomposition of contrastive learning, the numerator of Eq. (5) promotes *alignment* of positives while the log-sum-exp denominator promotes *uniformity* of the embedding distribution on the sphere, which is maximised only by a spread-out configuration and is therefore directly opposed to collapse. The joint objective (10) thus realises a controlled balance: $\mathcal{L}_{\text{scat}}$ pulls each normal class towards its centroid, $\mathcal{L}_{\text{con}}$ prevents the normal embeddings from degenerating to a point and keeps different samples separable, and λ governs the equilibrium. The unit-sphere constraint additionally bounds $\|\boldsymbol{\mu}_k\|_2 \leq 1$, so the penalty cannot diverge.

1.6. Anomaly scoring via linear one-class SVM

After training, we discard the contrastive heads’ interaction and retain the *concatenated* representation

$$\mathbf{z}_i = [\mathbf{h}_{1,i}^\top, \mathbf{h}_{2,i}^\top, \dots, \mathbf{h}_{K,i}^\top]^\top \in \mathbb{R}^{Kd}, \quad (11)$$

which fuses all K kernel views into a single descriptor. A linear-kernel OC-SVM is fitted on the normal embeddings $\{\mathbf{z}_i : y_i = 0\}$ by solving

$$\min_{\mathbf{w}, \rho, \xi} \quad \frac{1}{2} \|\mathbf{w}\|_2^2 - \rho + \frac{1}{\nu N_n} \sum_{i: y_i=0} \xi_i \quad \text{s.t.} \quad \langle \mathbf{w}, \mathbf{z}_i \rangle \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad (12)$$

where $\nu \in (0, 1]$ upper-bounds the fraction of normal training points allowed to fall outside the decision region. The anomaly score is the negative signed distance to the learned hyperplane,

$$a(\mathbf{x}_i) = \rho - \langle \mathbf{w}, \mathbf{z}_i \rangle, \quad (13)$$

so that points lying on the far (origin) side of the hyperplane receive high scores. We retain the *linear* kernel deliberately: the contrastive and scatter objectives jointly mould the normal class into a compact, nearly linearly separable cone on the product of spheres $(\mathbb{S}^{d-1})^K$, for which a linear separator is both sufficient and substantially cheaper than an RBF OC-SVM.

1.7. Overall algorithm and complexity

Algorithm 1 summarises the complete training and scoring pipeline. The dominant per-mini-batch cost is the cross-kernel contrastive term, $O(K^2 B^2 d)$ for the $\binom{K}{2}$ pairwise affinity matrices, while the scatter regularizer contributes only $O(K B d)$. Embedding extraction over the full data set is $O(N K d D)$, and fitting the linear OC-SVM scales with the number of support vectors. With a fixed small kernel bank ($K = 5$), the method retains the same asymptotic complexity as the contrastive backbone, so the accuracy gains reported in the experiments come at no additional order-of-growth cost.

References

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Algorithm 1 SCMK for one-class anomaly detection

Require: normal set $\mathcal{X}_n$, full set $\mathcal{X}$, kernel ratios $\{r_k\}_{k=1}^K$, embedding dim. d , weight λ , batch size B , epochs T , OC-SVM parameter ν

Ensure: anomaly scores $\{a(\mathbf{x}_i)\}_{i=1}^N$

- 1: $\text{med} \leftarrow$ median pairwise distance on $\mathcal{X}_n$ $\triangleright$ Eq. (1)
- 2: build kernels κ_k with $t_k = r_k \text{med}$ $\triangleright$ Eq. (2)
- 3: initialise projection heads $\{\mathbf{W}_k\}_{k=1}^K$
- 4: **for** epoch = 1 **to** T **do**
- 5: **for** each mini-batch of B samples from $\mathcal{X}_n$ **do**
- 6: $\mathbf{h}_{k,i} \leftarrow \mathbf{W}_k \mathbf{x}_i / \|\mathbf{W}_k \mathbf{x}_i\|_2$ for all k, i $\triangleright$ Eq. (3)
- 7: $\mathcal{L}_{\text{con}} \leftarrow$ cross-kernel InfoNCE over kernel pairs $\triangleright$ Eqs. (4)–(6)
- 8: $\mathcal{L}_{\text{scat}} \leftarrow -\frac{1}{K} \sum_k \left\| \frac{1}{B} \sum_i \mathbf{h}_{k,i} \right\|_2^2$ $\triangleright$ Eq. (7)
- 9: update $\{\mathbf{W}_k\}$ by a gradient step on $\mathcal{L}_{\text{con}} + \lambda \mathcal{L}_{\text{scat}}$ $\triangleright$ Eq. (10)
- 10: **end for**
- 11: **end for**
- 12: $\mathbf{z}_i \leftarrow [\mathbf{h}_{1,i}; \dots; \mathbf{h}_{K,i}]$ for all $\mathbf{x}_i \in \mathcal{X}$ $\triangleright$ Eq. (11)
- 13: fit linear OC-SVM on $\{\mathbf{z}_i : y_i = 0\}$ with parameter ν $\triangleright$ Eq. (12)
- 14: **return** $a(\mathbf{x}_i) = \rho - \langle \mathbf{w}, \mathbf{z}_i \rangle$ $\triangleright$ Eq. (13)

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